CSCI E-106:Assignment 2

Due Date: February 13, 2026 at 11:59 pm EST

Instructions

Students should submit their reports on Canvas. The report needs to clearly state what question is being solved, step-by-step walk-through solutions, and final answers clearly indicated. Please solve by hand where appropriate.

Please submit two files: (1) a R Markdown file (.Rmd extension) and (2) a PDF document, word, or html generated using knitr for the .Rmd file submitted in (1) where appropriate. Please, use RStudio Cloud for your solutions.

Problem 1

Refer to the CDI data set. Use the number of active physicians as Y and total personal income as X. (50 points)

a-) Built a regression model to predict Y by using X. Write down the stand error of the slope and intercept. (5 points)

b-) Are the slope and intercept statistically significant? Write down your null and alternative hypothesis and p value of the test. Calculate the 95% Confidence interval for the slope. (20 points)

c-) Predict the number of active physicians for total personal income of 100000,250000 and 50000. Calculate the 95% confidence interval (Hint: Use the range of total personel income to identify which observations are within the range of the training data) (20 points)

d-) What is the error variance of the regression line?(5 points)

Problem 2

Refer to the CDI data set. Use the number of active physicians as Y and total personal income as X. Select 1,000 random samples of 400 observations, fit the regression model and record β_0 and β_1 for each selected sample. Calculate the mean and variance of β_0 and β_1 based on the 1,000 different regression line and compare against the regression model in question 1 part b. (50 points)